Integration: Partial Fractions

Questions 1–14

Question 1. It is given that $f(x) = \dfrac{8x + 23}{x + 2}$.

a) Write $f(x)$ in the form $A + \dfrac{B}{x + 2}$.

b) Hence find $\int f(x)\,dx$.

Question 2. It is given that $f(x) = \dfrac{2x^2 - 2x - 1}{x + 1}$.

a) Write $f(x)$ in the form $Ax + B + \dfrac{C}{x + 1}$.

b) Hence find $\int f(x)\,dx$.

Question 3.

a) Express $\dfrac{4}{(x-3)(x+1)}$ as partial fractions.

b) Show that $\displaystyle\int \frac{4}{(x-3)(x+1)}\,dx = \ln\left|\frac{x-3}{x+1}\right| + c$.

Question 4.

a) Find $\displaystyle\int \frac{10}{25-x^2}\,dx$.

b) Hence show that $\displaystyle\int_{-1}^{1} \frac{10}{25-x^2}\,dx = \ln\frac{a}{b}$ where a and b are integers to be found.

Question 5.

a) Find $\displaystyle\int \frac{2x-1}{(x+5)(x-6)}\,dx$.

b) Hence find the value of k such that $\displaystyle\int_7^k \frac{2x-1}{(x+5)(x-6)}\,dx = \ln 5$.

Question 6.

a) Express $\dfrac{k-4x}{x(2-x)}$ as partial fractions, where k is a positive constant.

b) Find the value of k such that $\displaystyle\int_3^4 \frac{k-4x}{x(2-x)}\,dx = \ln\frac{128}{27}$.

Question 7.

a) Express $\dfrac{23x+11}{(3-x)(x+2)(3x-1)}$ as partial fractions.

b) Write $\displaystyle\int_1^2 \frac{23x+11}{(3-x)(x+2)(3x-1)}\,dx$ in the form $\ln k$, where k is a positive constant.

Question 8.

a) Express $\dfrac{4x^2+1}{x^2(2x+1)}$ as partial fractions.

b) Show that $\displaystyle\int \frac{4x^2+1}{x^2(2x+1)}\,dx = \frac{a}{x} + \ln f(x) + c$ where a is an integer to be found.

Question 9.

a) Write $\dfrac{4x-2}{(x-2)^2}$ in the form $\dfrac{A}{x-2}+\dfrac{B}{(x-2)^2}$ where A and B are constants.

b) Show that $\displaystyle\int_3^4 \frac{4x-2}{(x-2)^2}\,dx = a+\ln b$ where a and b are integers to be found.

Question 10.

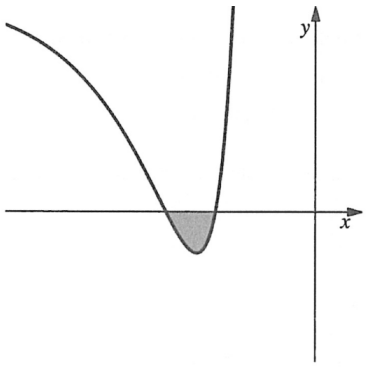
a) Find $\dfrac{10x^2+6}{(x+1)^2(3x-1)}$ as partial fractions.

b) Hence show that $\displaystyle\int_1^{5/3} \frac{10x^2+6}{(x+1)^2(3x-1)}\,dx = a+b\ln 2+c\ln 9$ where a, b and c are constants to be found.

Question 11. The diagram below shows the curve with equation

$$y = \frac{x^2 + 5x + 6}{x^2(2 - x)}, \qquad x \neq 0,\ x \neq 2.$$

Find the exact value of the shaded area.



Question 12. Find the exact value of

$$\lim_{\delta x \to 0} \sum_{x=0}^{1/8} \frac{360x^2 + 157x + 16}{(3x + 1)(2 - 5x)(8x + 3)}\, \delta x.$$

Question 13. Prerequisite knowledge for Q13–Q14: Integration by Substitution Use the substitution $u = \sqrt{x} - 2$ to find

$$\int \frac{9}{2(\sqrt{x}-2)(\sqrt{x}+1)(\sqrt{x}+4)}\,dx.$$

Question 14. Use the substitution $u = \sqrt{x-2}$ to show that

$$\int_3^6 \frac{6}{(3+2\sqrt{x-2})(x-2)}\,dx = 4\ln\frac{10}{7}.$$